Swiss Institute of Bioinformatics

INTRODUCTION TO SEQUENCING DATA ANALYSIS

Reproducible Computational Research

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Adapted from previous year courses

elixir SWITZERLAND

Learning Objectives

Understand the difference between replicable and reproducible research.

Learn about the reproducibility crisis and its impact on science.

Get introduced to literate programming for clearer, more reproducible code.

Apply simple rules to make your own code reproducible and easy to understand.

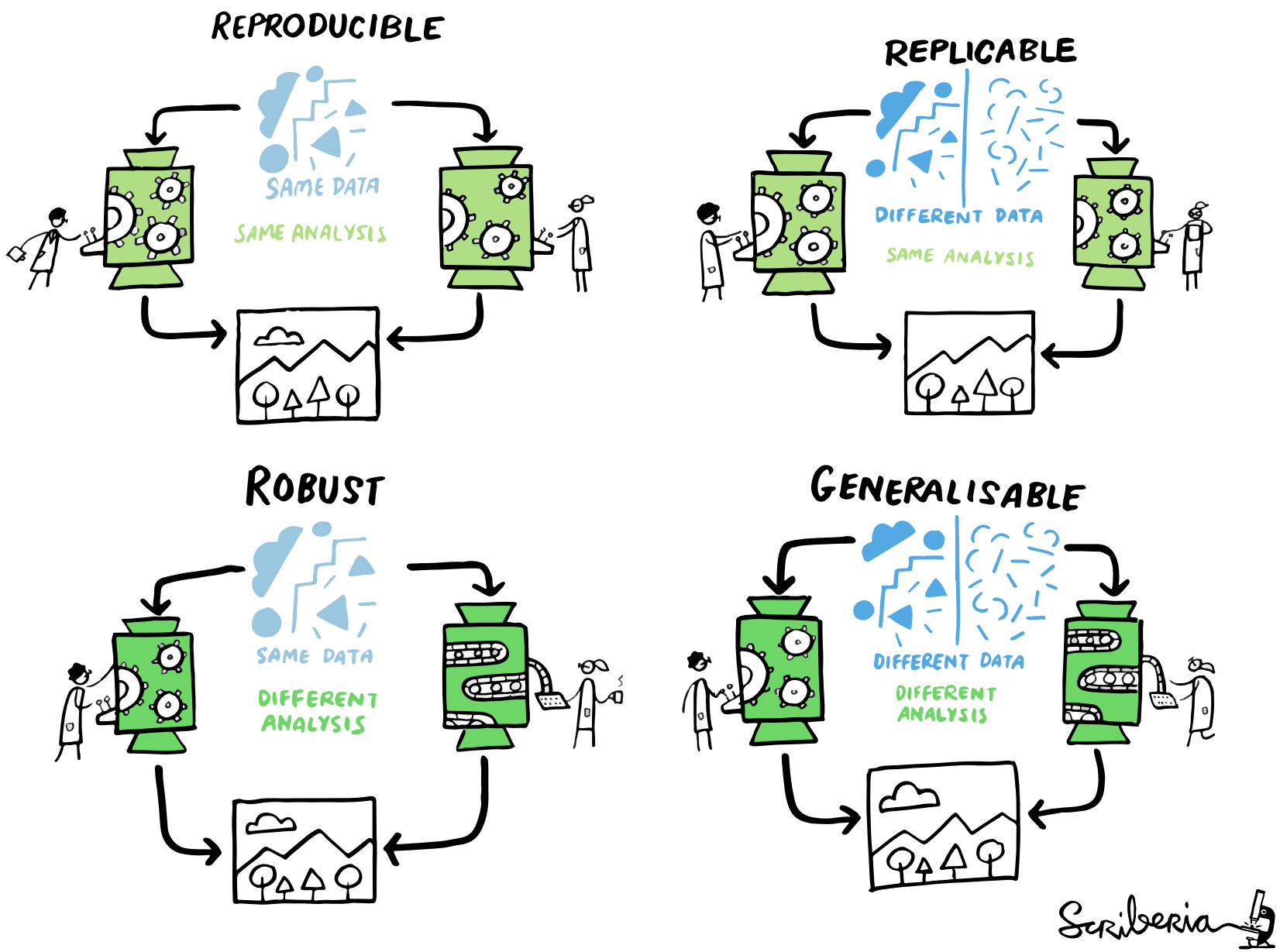
Computational reproducibility is the ability to **recreate**, **exactly same**, an earlier **research/ analysis** given the **same data** and **code**.

Replication vs Reproducibility

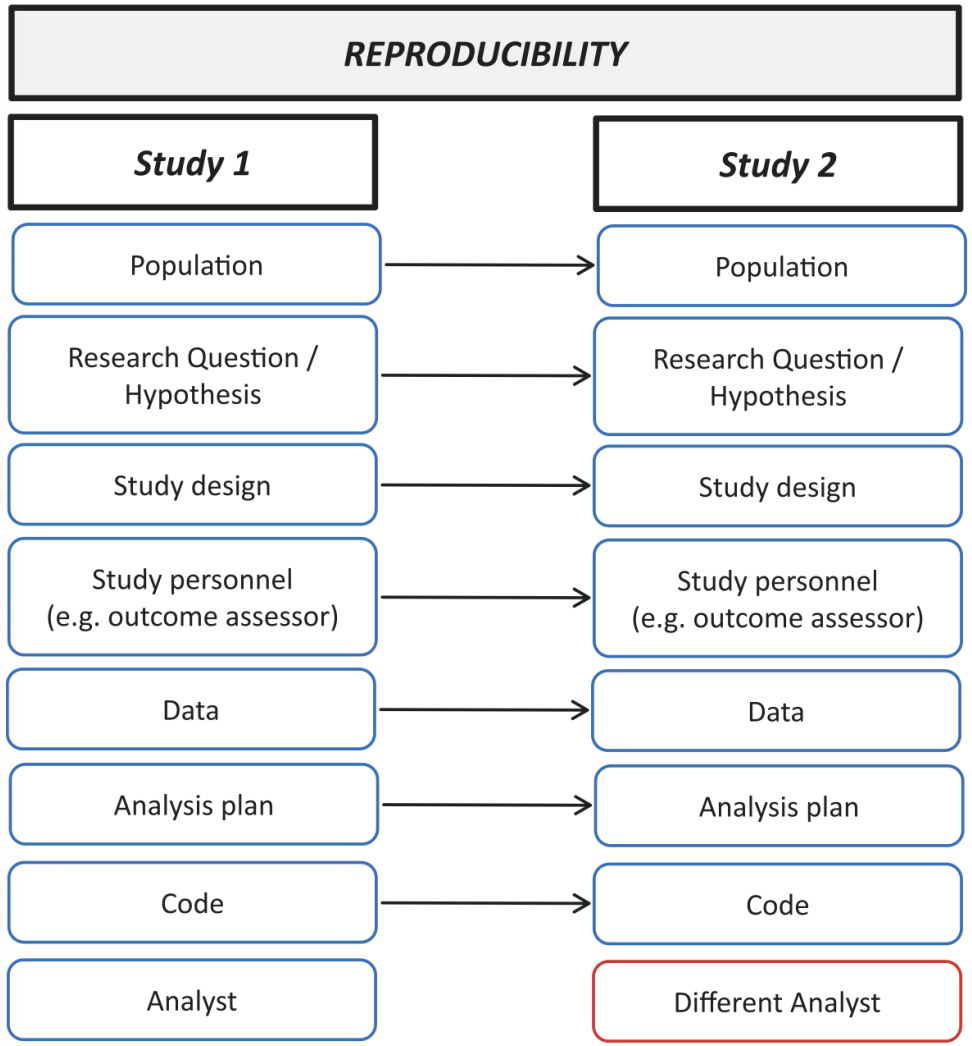
		Data	
		Same	Different
Analysis	Same	Reproducible	Replicable
	Different	Robust	Generalisable

Claerbout, J. F., & Karrenbach, M. (1992). Electronic documents give reproducible research a new meaning. In *SEG Technical Program Expanded Abstracts*. https://doi.org/10.1190/1.1822162

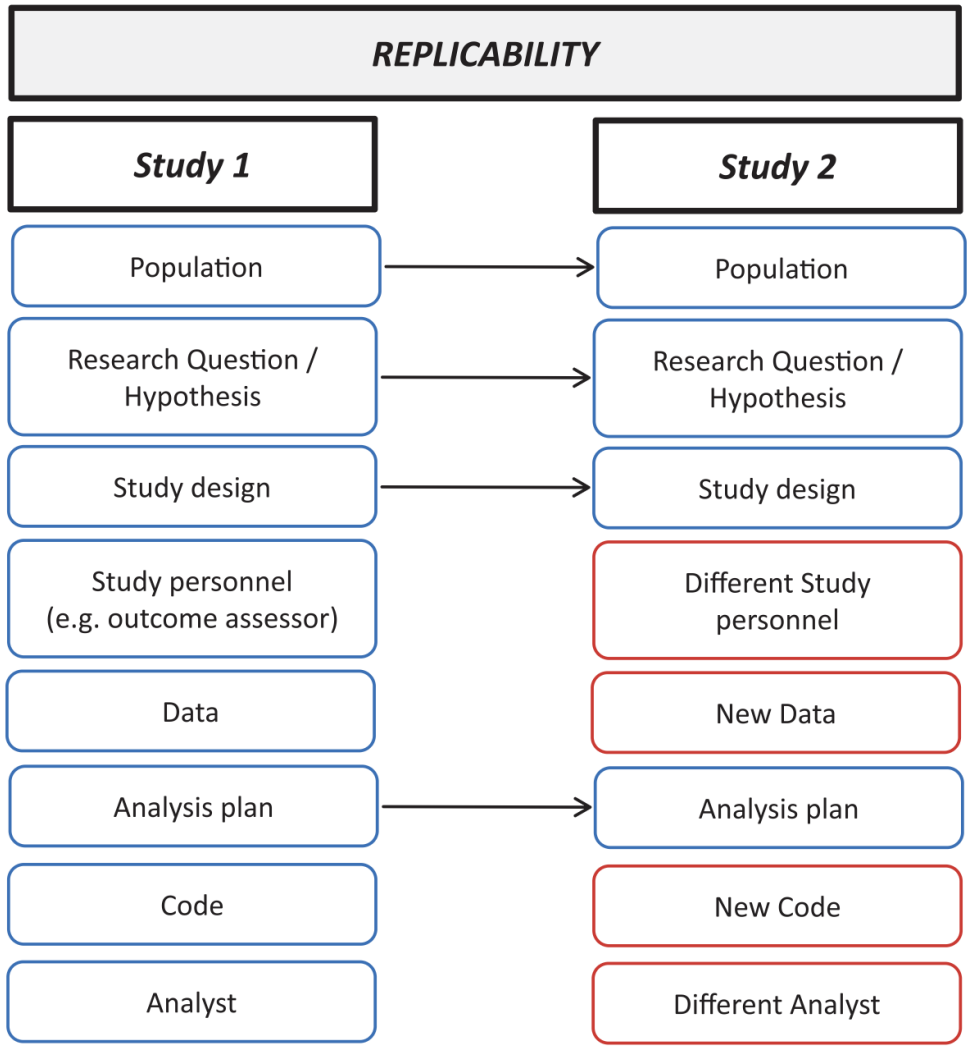
Cartoon depiction of terms



Replicability vs Reproducibility



Study 2 has successfully ***reproduced*** Study 1 if the estimates from both studies are consistent



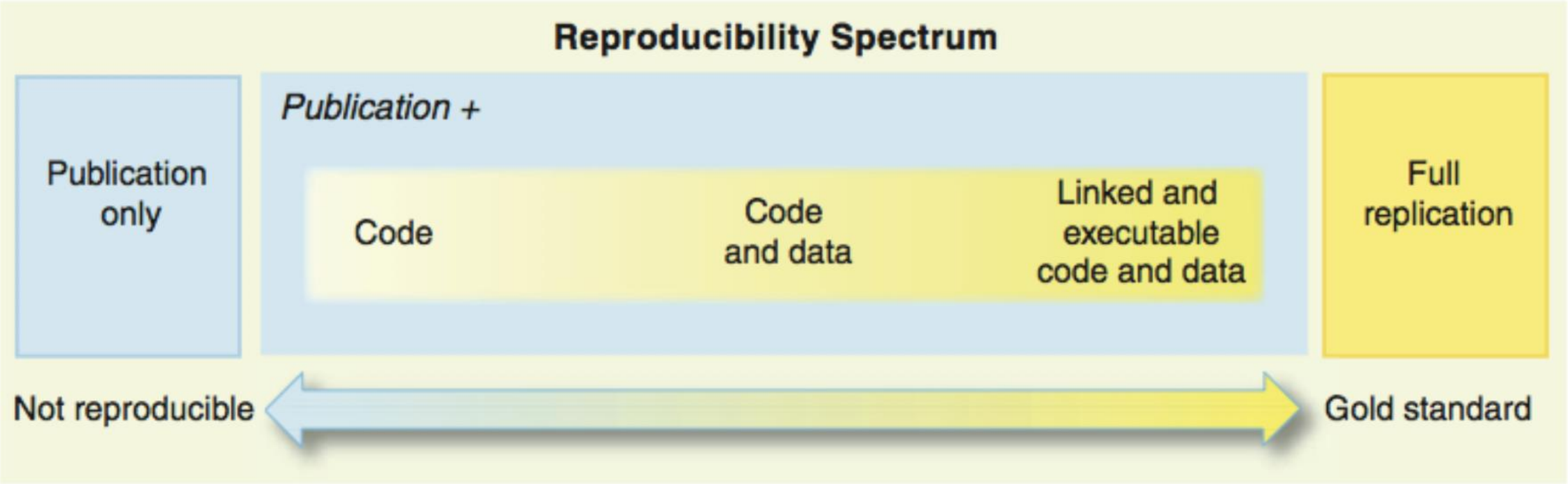
Study 2 has successfully ***replicated*** Study 1 if the estimates from both studies are consistent

10.1097/j.pain.0000000000001254

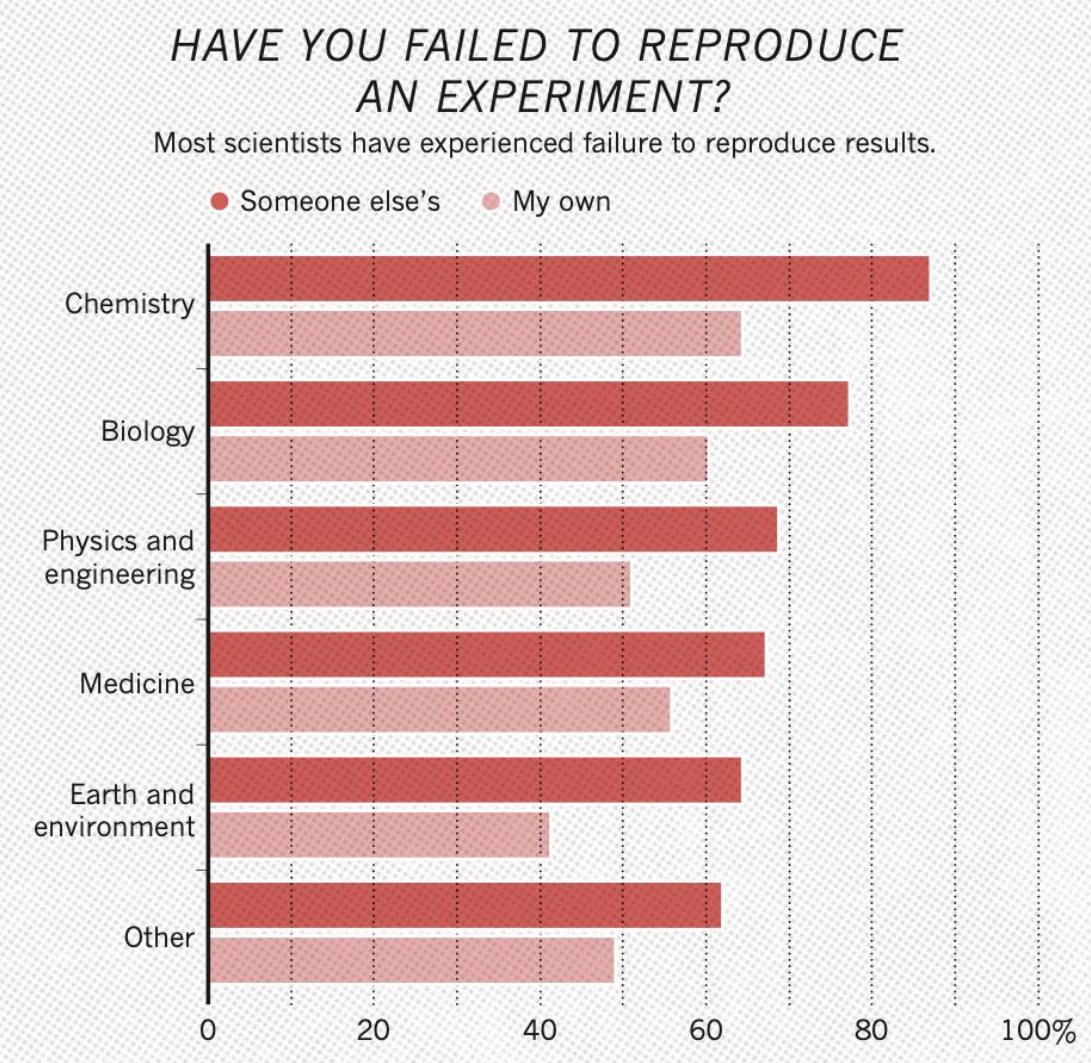
Quiz: 8

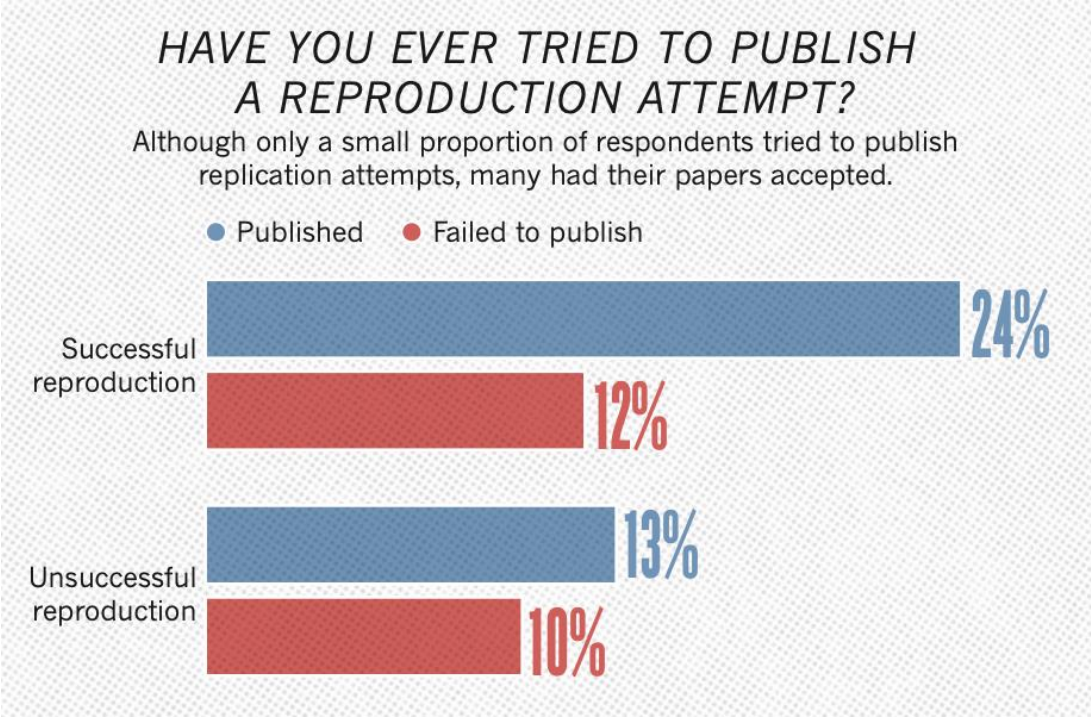
What best describes *computational reproducibility*?

Reproducibility spectrum



The reproducibility crisis in science





https://www.nature.com/articles/533452a

Number of respondents from each discipline:
Biology **703**, Chemistry **106**, Earth and environmental **95**, Medicine **203**, Physics and engineering **236**, Other **233**

Quiz: 9

What is one key issue highlighted in the reproducibility crisis?

Academic Bias and biotech failures

The unspoken fact is that at least 50% of the studies published even in top tier academic journals – Science, Nature, Cell, PNAS, etc… - can't be repeated with the same conclusions by an industrial lab. In particular, key animal models often don't reproduce.

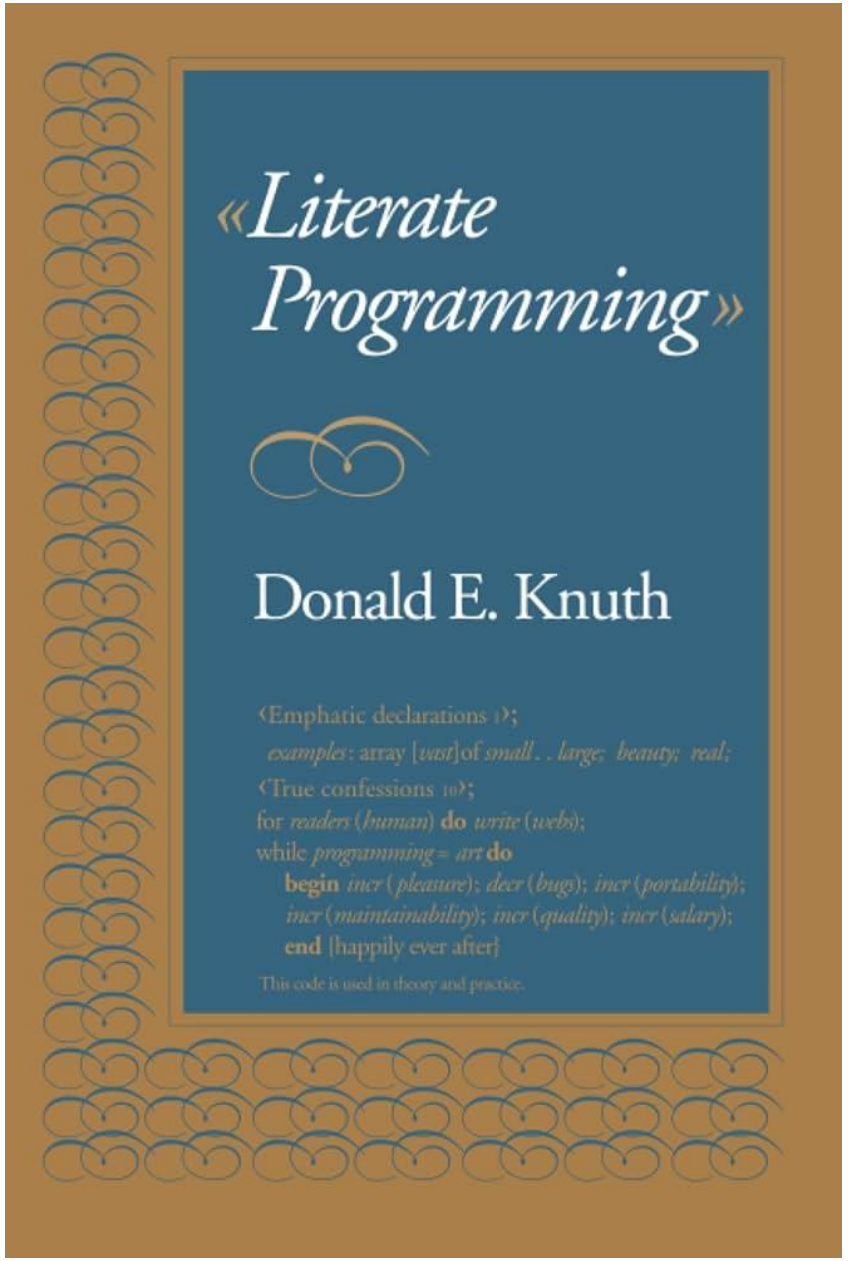
https://lifescivc.com/2011/03/academic-bias-biotech-failures/

Incipit liber geneseos quē nos greci dicimus

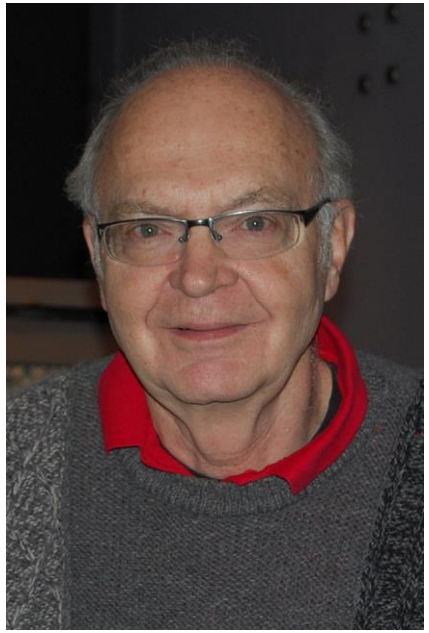
In principio creauit deus celū. et terram. Terra autem erat inanis et
vacua: et tenebre erant super faciē abissi:
et spiritus dei ferebatur super aquas.
Dixitq; deus. Fiat lux. Et facta ē lux.
Et vidit deus lucem q̄ esset bona: et
diuisit lucem a tenebris. appellauitq;
lucem diem et tenebras noctem. factū
q; est vespere et mane dies unus. Dixit
quoq; deus. Fiat firmamentū in me-
dio aquarū: et diuidat aquas ab a-
quis. Et fecit deus firmamentū: diui-
sitq; aquas que erant sub firmamen-
to ab his que erant super firmamen-
tum: et factum est ita. Vocauitq; deus
firmamentū celū: et factum est vespere
et mane dies secundus. Dixit vero de-
us. Congregentur aque que sub celo
sunt in locum unū: et appareat arida.
Et factum est ita. Et vocauit deus ari-
dam terram: cōgregationesq; aquarū
appellauit maria. Et vidit deus q̄ es-
set bonū. et ait. Germinet terra herbā
virentem et facientem semen: et lignū
pomiferū faciens fructum iuxta genus
suū: cuius semen in semetipso sit super
terram. Et factum est ita. Et protulit
terra herbam virentem et afferentem se-
men iuxta genus suū: ligniq; faciens
fructū et habēs unūq̄q; semen secūm
speciē suā. Et vidit deus q̄ esset bonū:
et factū ē vespere et mane dies tertius.
Dixitq; aūt deus. Fiant luminaria
in firmamēto celi. et diuidāt diem ac
noctē: et sint in signa et tēpora. et dies et
annos: ut luceāt in firmamēto celi et
illuminēt terrā. Et factū est ita. Fecitq;
deus duo luminaria magna: lumina-
ria ut p̄esset diei et luminare min-
us ut p̄esset nocti: et stellas. et posuit eas in
firmamēto celi ut luceret super terrā: et
p̄esset diei ac nocti: et diuidere lucem
ac tenebras. Et vidit deus q̄ esset bonū:
et factū ē vespere et mane dies quartus.
Dixit etiam deus. Producant aque
repile anime viuentis et volatile super
terram: sub firmamēto celi. Creauitq;
deus cete grandia. et omnē animā vi-
uentem atq; motabilem quā produxe-
rant aque in species suas: et omne vo-
latile secundū genus suū. Et vidit de-
us q̄ esset bonū: benedixitq; ei dicens.
Crescite et multiplicamini. et replete a-
quas maris: auesq; multiplicentur
super terram. Et factū ē vespere et mane
dies quintus. Dixit quoq; deus. Pro-
ducat terra animā viuentem in gene-
re suo: iumenta et reptilia. et bestias ter-
re secundū species suas. Factū ē ita. Et
fecit deus bestias terre iuxta species su-
as: iumenta et omne repile terre in ge-
nere suo. Et vidit deus q̄ esset bonū:
et ait. Faciamus hominē ad ymaginē et
similitudinē nostrā. et p̄sit pisibus maris.
et volatilibus celi. et bestijs. uniūseq; terre:
omniq; reptili q̄ mouet ī terra. Et crea-
uit deus hominē ad ymaginē et simi-
litudinē suam: ad ymaginem dei creā-
uit illū: masculū et feminā creauit eos.
Benedixitq; illis deus. et ait. Crescite
et multiplicamini et replete terram. et
subiicite eam: et dominamini pisibus
maris. et volatilibus celi: et universis
animantibus que mouentur super terrā.
Dixitq; deus. Ecce dedi vobis omnē
herbam afferentem semen super terram.
et universa ligna que habent ī semetipsis
sementē generis sui: ut sint vobis ī escā.
et cūctis aīantibus terre. omniq; volucri
celi et universis q̄ mouentur in terra. et ī
quibus ē anima viuēs: ut habeāt ad
vescendū. Et factū est ita. Vidit q̄ deus
cuncta que fecerat: et erāt valde bona.

A perfectly typeset text example

Literate programming



Literate programming is a programming methodology that combines a programming language with a documentation language, making programs more robust, more portable, and more easily maintained than programs written only in a high-level language.



Quiz: 10

What is the goal of literate programming?

Tools for literate programming in R

rmarkdown

quarto®

Tools for literate programming in LaTeX

Sweave – Integrates R code into LaTeX (older tool).

knitr – Improved version of Sweave, works with R and LaTeX.

Pweave – Python + LaTeX (like knitr, but for Python).

noweb – Original language-independent tool for literate programming.

Tools for literate programming in Multiple-languages

quarto®

Org Mode

Your life in plain text

A *GNU Emacs* major mode for keeping notes, authoring documents, computational notebooks, literate programming, maintaining to-do lists, planning projects, and more — in a fast and effective plain text system.

GNU ELPA | org 9.7.28 | report bug | feedback | source code | sr.ht Project

Libera.Chat | Matrix

Quiz: 11

Which of the following is not a tool for literate programming?

Why reproducible?

1. For yourself!
   a. Adjusting your analysis
   b. Sharing your analysis
   c. Find out what the heck you did > 2 weeks ago

2. Because the academic community requires it..
   1. Many journals require accompanied code
   2. Proposals often require a data management plan

5 simple rules to get started

1. Execute the commands from a script
2. Number scripts based on their order of execution
3. Give your scripts a descriptive and active name
4. Make your scripts specific
5. Directories and variables at the beginning of the script

Rule 1

Execute the commands from a script to be able to trace back your steps

All output **files** and **directories** created from within a script

Adjusting your analysis becomes possible

Makes your analysis **portable**. It can be run:

- On a different computer
- By your colleague

Rule 2

Number scripts based on their order of execution (e.g. 01_download_reads.sh)

Easily trace the order of execution

Seperates **main** scripts from **secondary** scripts (i.e. scripts called by another script)

Example:
01_download_reads.sh
02_run_fastqc.sh
03_trim_reads.sh
04_run_fastqc_trimmed.sh

Rule 3

Give your scripts a **descriptive and active names**

Makes it easier to identify the script of interest

Example:

01_download_reads.sh
02_run_fastqc.sh
03_trim_reads.sh
04_run_fastqc_trimmed.sh

Rule 4

Make your scripts **specific**, i.e. do not combine many different commands in the same script

Makes your scripts **modular** (i.e. you can use it for other analyses)

Makes **job submission** more efficient

Turn your script in a **pipeline** later

Rule 5

Refer to **directories and variables at the beginning** of the script

Directories and variables need to be **changed** often
No need to **search** through the whole script to change them
Debugging is easier

Example:

```
#!/usr/bin/env bash
TRIMMED_DIR=~/workdir/trimmed_data
READS_DIR=~/workdir/reads
mkdir -p $TRIMMED_DIR
```

Quiz: 12

Which of the following is not one of the five simple rules for reproducible code?

Further steps

Version control (git, GitHub, GitLab)

Pipelines

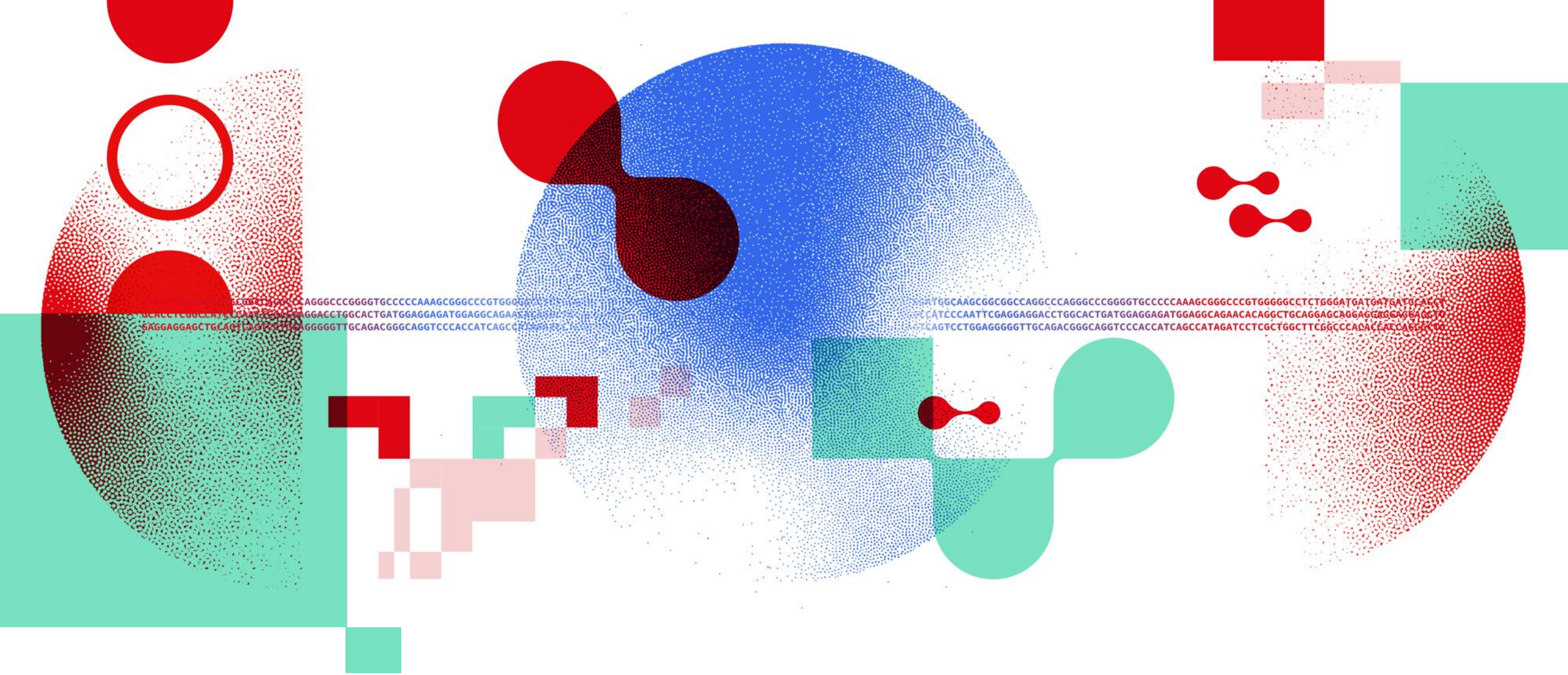
Notebooks

Further readings

1. Sandve, G.K. *et al.* (2013b) 'Ten simple rules for reproducible computational research,' *PLoS Computational Biology*, 9(10), p. e1003285. https://doi.org/10.1371/journal.pcbi.1003285.
2. Heise, V. *et al.* (2023) 'Ten simple rules for implementing open and reproducible research practices after attending a training course,' *PLoS Computational Biology*, 19(1), p. e1010750. https://doi.org/10.1371/journal.pcbi.1010750.
3. Rule, A. *et al.* (2018b) 'Ten simple rules for reproducible research in Jupyter notebooks,' *arXiv (Cornell University)* [Preprint]. https://doi.org/10.48550/arxiv.1810.08055.

Summary

1. **Reproducibility is essential** for trust and progress in science.
2. **Reproducibility ≠ Replicability** – know the difference.
3. The **reproducibility crisis** affects even top-tier research.
4. **Literate programming** improves transparency by combining code and explanation.
5. Follow **simple coding rules** to make your work reproducible.
6. Think ahead: **version control, documentation, and sharing** are key for reproducible research.



Thank you

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